

# Design, Evaluation, and Implementation of Synthetic Isopentyldiol Pathways in *Escherichia coli*

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Cite This: *ACS Synth. Biol.* 2023, 12, 3381–3392



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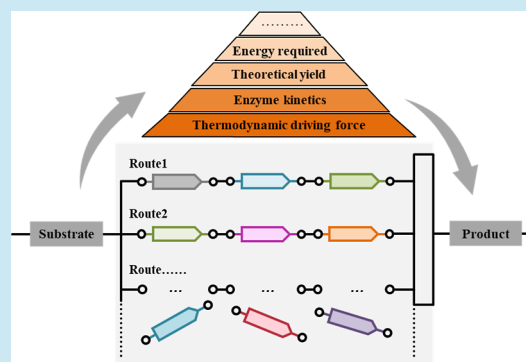
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**ABSTRACT:** Isopentyldiol (IPDO) is an important raw material in the cosmetic industry. So far, IPDO is exclusively produced through chemical synthesis. Growing interest in natural personal care products has inspired the quest to develop a biobased process. We previously reported a biosynthetic route that produces IPDO via extending the leucine catabolism (route A), the efficiency of which, however, is not satisfactory. To address this issue, we computationally designed a novel non-natural IPDO synthesis pathway (route B) using RetroPath RL, the state-of-the-art tool for bioretrosynthesis based on artificial intelligence methods. We compared this new pathway with route A and two other intuitively designed routes for IPDO biosynthesis from various perspectives. Route B, which exhibits the highest thermodynamic driving force, least non-native reaction steps, and lowest energy requirements, appeared to hold the greatest potential for IPDO production. All three newly designed routes were then implemented in the *Escherichia coli* BL21(DE3) strain. Results show that the computationally designed route B can produce 2.2 mg/L IPDO from glucose but no IPDO production from routes C and D. These results highlight the importance and usefulness of in silico design and comprehensive evaluation of the potential efficiencies of candidate pathways in constructing novel non-natural pathways for the production of biochemicals.

**KEYWORDS:** isopentyldiol, diol, retrosynthesis, pathway evaluation, thermodynamics analysis, synthetic biology



## 1. INTRODUCTION

Presently, the majority of value-added compounds, such as solvents, biofuels, and polymer precursors, are produced through petroleum-dependent and energy-intensive chemical processes. Great concerns about climate change, air pollution, and oil price fluctuation have inspired the quest for alternative green processes that produce the desired products with low toxic byproduct formation and carbon dioxide emission. Biobased production processes using renewable feedstocks as substrates hold great potential for a sustainable supply of chemicals. Synthetic biology aims at designing and altering the host metabolic system as well as introducing artificial pathways whose behavior can be predicted, programed, and, finally, controlled.<sup>1–3</sup>

Short-chain (C3–C5) diols are considered a class of important green platform chemicals that have wide applications in manufacturing a variety of commercial chemical products, such as cosmetics, feed additives, polymers, solvents and pharmaceuticals.<sup>4–6</sup> Isopentyldiol (IPDO) is an appealing ingredient in cosmetic industry. It has no odor or irritation and is evaluated as a safe and eco-friendly material. Compared to other diols used in cosmetics (such as propylene glycol, butylene glycol, or dipropylene glycol), IPDO presents better skin feel, and deodorant and antibacterial performances and

therefore can work as a moisturizer, solubilizer, and booster of preservatives. Recently, we proposed a general route for microbial production of structurally diverse diols by expanding the amino acid metabolism.<sup>7</sup> This general route combines oxidative and reductive formations of the diol hydroxyl groups and comprises four reactions catalyzed by amino acid hydroxylase, L-amino acid deaminase,  $\alpha$ -keto acid decarboxylase, and aldehyde reductase consecutively. As a proof of concept, the feasibility of IPDO production via this general route has been verified (named here route A). To increase the IPDO titer, further efforts have been made, including screening for pathway enzymes with higher activity, enhancing the supply of the precursor amino acid, and directed evolution of the key enzyme hydroxylase. The achieved IPDO titer of 310 mg/L is, however, still far from the requirement of a practical application. In view of this, we explore in this study the

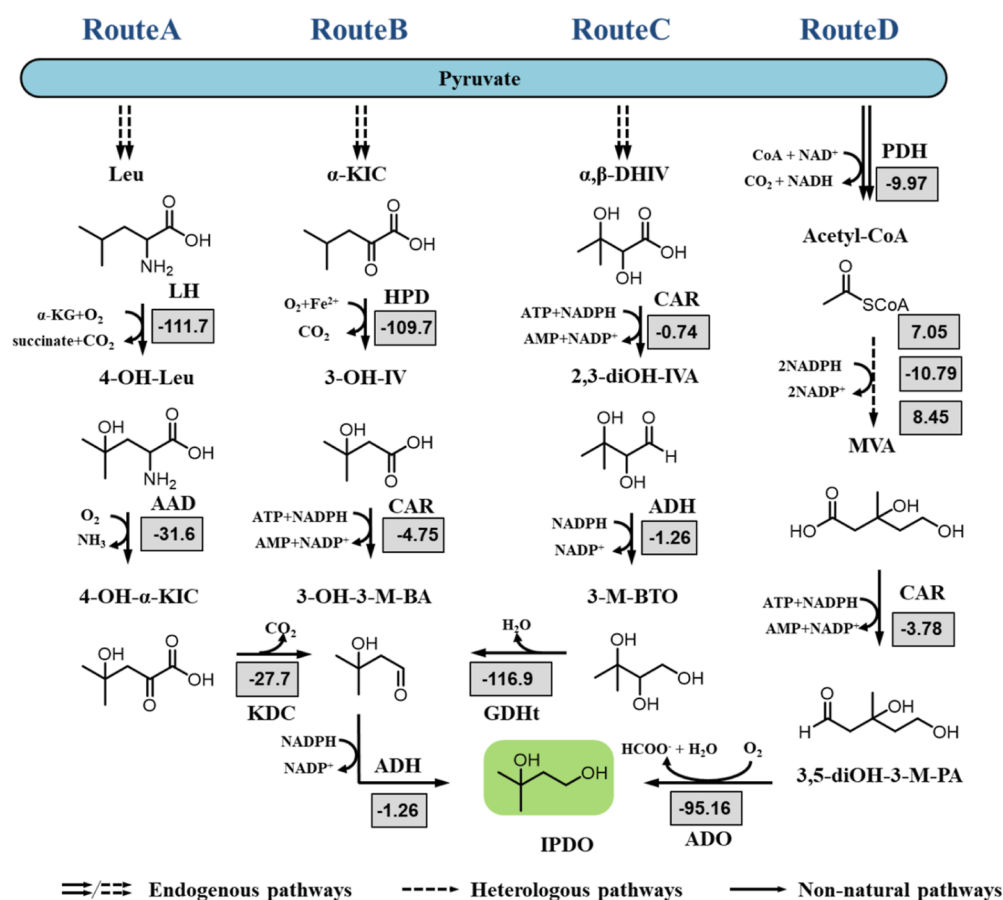
**Received:** June 29, 2023

**Revised:** September 13, 2023

**Accepted:** October 10, 2023

**Published:** October 23, 2023





**Figure 1.** Proposed synthetic routes for IPDO production. All four routes produce IPDO by extending the endogenous intermediate metabolism of *E. coli*. Routes A and route B divert flux from the leucine synthesis pathway and are composed of four and three exogenous reaction steps, respectively. Route C starts from  $\alpha,\beta$ -DHIV and consists of four non-native reaction steps. In route D, IPDO is synthesized by extending the MVA pathway via two exogenous reaction steps. The standard Gibbs free formation energy of every non-natural product was estimated using the group contribution method.<sup>32</sup> Abbreviations: LH, leucine hydroxylase; AAD, L-amino acid deaminase; KDC,  $\alpha$ -keto decarboxylase; ADH, alcohol dehydrogenase; HPD,  $\alpha$ -ketoisocaproate dioxygenase; CAR, carboxylic acid reductase; GDHt, glycerol dehydratase; ADO, aldehyde-deformylating oxygenase; 4-OH-Leu, 4-hydroxyleucine; 4-OH- $\alpha$ -KIC, 4-hydroxy- $\alpha$ -ketoisocaproate; 3-OH-3-M-BA, 3-hydroxy-3-methylbutanal;  $\alpha$ -KIC,  $\alpha$ -ketoisocaproate; 3-OH-IV, 3-hydroxyisovalerate;  $\alpha,\beta$ -DHIV,  $\alpha,\beta$ -dihydroxyisovalerate; 2,3-diOH-IVA, 2,3-dihydroxy-3-methylbutanal; 3-M-BTO, 3-methyl-1,2,3-butanetriol; Pyr, pyruvate; MVA, mevalonate; 3,5-diOH-3-M-PA, 3,5-dihydroxy-3-methylpentanal; IPDO, isopentylidol.

possibility of producing IPDO via alternative artificial biosynthetic routes.

Designing an artificial biosynthetic route is a process that generally consists of three steps: (1) design and evaluation of pathways; (2) verification of the designed pathways in vivo or in vitro; and (3) improving the production of target compounds by applying metabolic engineering strategies.<sup>2</sup> The development of computational pathway prediction tools has largely facilitated the identification of possible metabolic pathways and the corresponding enzymes. Currently, a number of prediction algorithms have been developed to aid in the biosynthetic route design, such as BNICE.ch,<sup>8</sup> RetroPath RL,<sup>9</sup> and Retrobiocat.<sup>3</sup> These pathway design tools use the retrosynthetic analysis (RTSA) method, often used in chemical synthesis. RTSA uses a set of reaction rules to backtrack from target products to potential reactants and connect individual reactions to form pathways. A scoring function is then used to rank the generated pathways based on reaction thermodynamic parameters, substrate/product similarity, potential toxicity of intermediates, and the number of enzymes that need to be heterologously expressed so as to select for more rational biosynthetic pathways. The advantage of the RTSA-based pathway design is that the target product is not limited to

known natural metabolites of a microbial host but can also be a completely new molecular structure. Moreover, as the technology of enzyme design advances, the diversity of substrates that enzymes can catalyze will broaden, enabling the synthesis of more diverse products using the generalized reaction rules of the retrosynthetic pathway design method. One landmark in the application of computer-aided design in metabolic engineering is the biological production of 1,4-butanediol.<sup>10</sup> In their work, more than 10,000 pathways were initially identified using the software Bio-Pathway Predictor, followed by selection using a pathway selection software to sort and rank the pathways based on attributes including maximum theoretical yield, thermodynamic feasibility, number of non-native reaction steps, and pathway length. As a result, two high-ranking artificial routes were selected for metabolic engineering, and the best recombinant strain produced 18 g/L 1,4-PDO from glucose after 5 days of cultivation.

Prior to the experimental implementation of the three designed routes, we investigated the potential of these designs by using two in silico analysis tools. For comparison, route A serves as the reference route and is evaluated with the other three routes in parallel. We used the flux balance analysis (FBA)<sup>11</sup> to predict the maximum IPDO yield and the max-

Table 1. Criteria for Evaluating the Potential Efficiencies of Four IPDO Synthetic Routes

| criteria                            | route A             |                                          | route B                                                                                                     | route C            | route D |
|-------------------------------------|---------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------|---------|
| theoretical yield (mol/mol glucose) | 0.628               | 0.667                                    | 0.827                                                                                                       | 0.667              |         |
| MDF values (kJ/mol)                 | 31.54 <sup>a</sup>  | 31.54 <sup>a</sup><br>84.67 <sup>b</sup> | 6.96 <sup>a</sup><br>74.93 <sup>b</sup>                                                                     | 38.37 <sup>a</sup> |         |
| NAD(P)H required                    | 2                   | 2                                        | 4                                                                                                           | 2                  |         |
| ATP required                        | 0                   | 1                                        | 1                                                                                                           | 1                  |         |
| number of non-native steps          | 4                   | 3                                        | 4                                                                                                           | 5                  |         |
| penalty                             | intermediate enzyme | $\alpha$ -KIC (low affinity to HPD)      | $\alpha,\beta$ -DHIV (low affinity to CAR)<br>GDHt (cofactor: VB <sub>12</sub> )<br>ADO (unclear mechanism) |                    |         |
| verification                        | yes                 | yes                                      | no                                                                                                          | no                 |         |

<sup>a</sup>Note: MDF values of all reactions in IPDO synthesis routes. <sup>b</sup>MDF values of the rest reactions after removing the last common rate-limiting reaction catalyzed by PDDH in routes A and B.

min driving force (MDF)<sup>12</sup> to determine the pathway thermodynamics so as to predict the feasibility and to identify the key bottlenecks in these pathways. Both tools are well-elucidated and can be easily implemented without the requirement of experimental data, thereby providing a simple and effective means of evaluating different artificial pathways that produce the same target compound. Afterward, in comparison with route A, all three artificial biosynthetic routes were separately introduced into the *Escherichia coli* BL21-(DE3) strain, and their performances on IPDO production were analyzed and compared with the in silico prediction results.

## 2. RESULTS

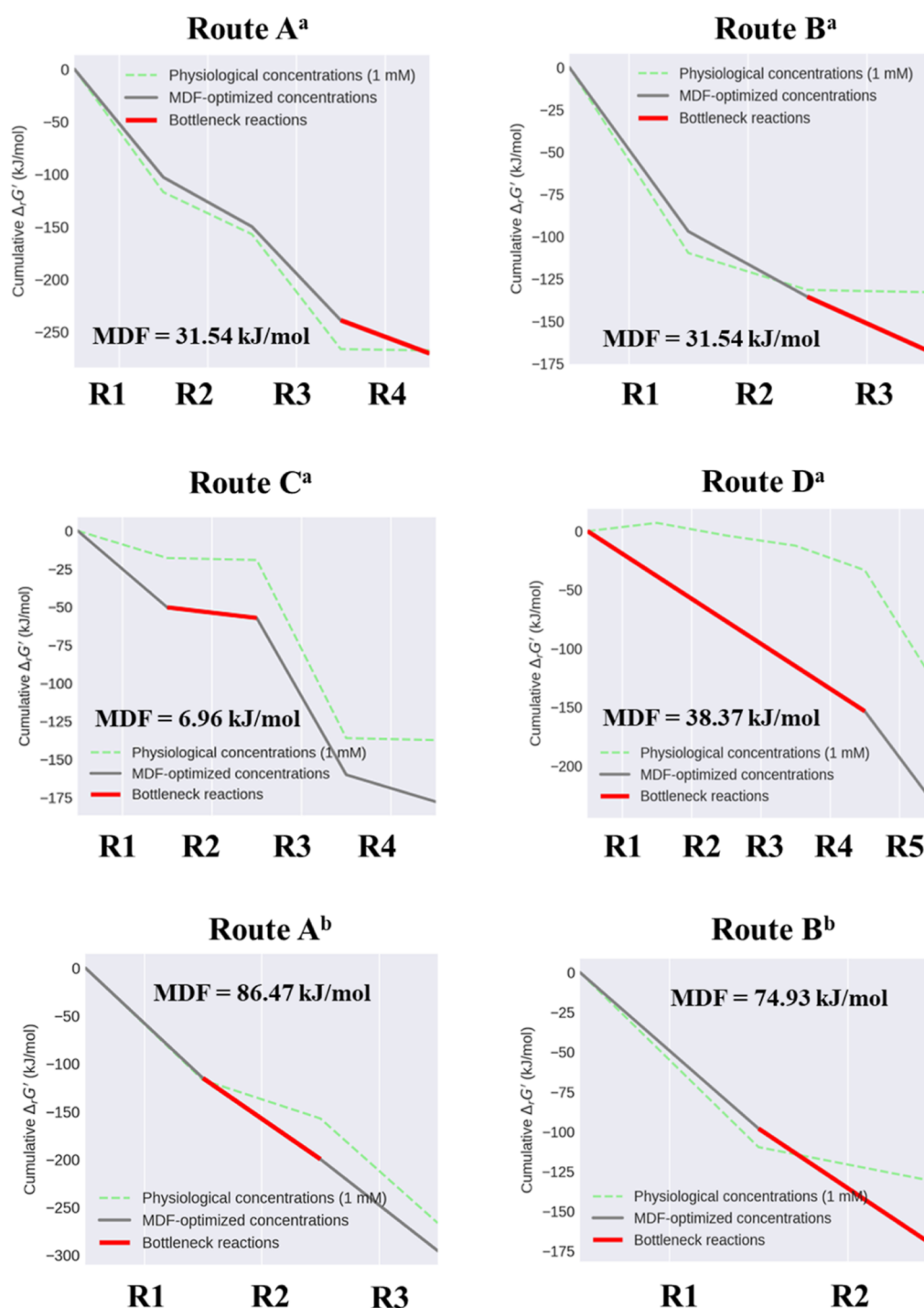
**2.1. Design and Evaluation of the Novel IPDO Biosynthetic Routes.** To find a potential retrosynthetic pathway, we used the bioretrosynthesis analysis software Retropath RL to make a computational prediction of the biosynthetic pathway for the target product IPDO. The prediction gave only one reasonable biosynthetic pathway: first,  $\alpha$ -KIC was oxidatively decarboxylated by 4-hydroxyphenylpyruvate dioxygenase/ $\alpha$ -ketoisocaproate dioxygenase (HPD) (EC: 1.13.11.27) to produce 3-OH-IV (Figure 1). 3-OH-IV was then converted to 3-hydroxy-3-methylbutanal (3-OH-3-M-BA) by a carboxylic acid reductase (CAR) (EC: 1.2.1.30). Subsequently, 3-OH-3-M-BA was reduced to IPDO by alcohol dehydrogenase (ADH) (EC: 1.1.1.202). Another pathway was obtained from the second round of selection after modification of the parameters (Figure S1). In this route, 3-OH-IV was obtained by hydrolysis of ethyl 3-hydroxyisovalerate (E-3-OH-IV), which was picked as the starting metabolite of the designed route. However, considering that this precursor is not a natural metabolic intermediate of *E. coli* but a structure deduced by retrosynthesis, we chose the former pathway for further research.

We notice that one distinct structural feature of IPDO is that it contains an isobutyl group, which motivates us to develop novel biosynthesis pathways that generate IPDO from metabolites that share a similar structure. In this regard, we propose two new biosynthetic routes (routes C and D) by selecting  $\alpha,\beta$ -dihydroxyisovalerate ( $\alpha,\beta$ -DHIV), an intermediate of the valine biosynthesis pathway, and mevalonate (MVA), respectively, as the precursors for IPDO production. Specifically, in route C, CAR catalyzes the reduction of  $\alpha,\beta$ -DHIV to generate 2,3-dihydroxyisovaleraldehyde (2,3-diOH-IVA), which in turn is converted to 3-methylbutane-1,2,3-triol (3-M-BTO) by PDDH. 3-M-BTO is transformed to 3-OH-3-M-BA by glycerol dehydratase (GDHt, EC: 4.2.1.30) and

subsequently to IPDO by PDDH. In route D, IPDO is produced by stretching the well-elucidated heterologous MVA catabolism.<sup>13,14</sup> Specifically, CAR is employed to reduce MVA to 3,5-dihydroxy-3-methylpentanal (3,5-diOH-3-M-PA), which is subsequently converted to IPDO by aldehyde-deformylating oxygenase (ADO) (EC: 4.1.99.5).

Unfortunately, the bioretrosynthesis analysis software Retropath RL is unable to forecast routes C and D, even with parameter adjustments. This stems from the rigidity of the rules enforced by Retropath RL concerning the nonreactive atoms neighboring the reaction center. This strict criterion results in the omission of metabolites with similar structures during the path exploration process within Retropath RL. As a result, the algorithm's capacity to uncover potential non-natural pathways becomes constrained.

Then, to evaluate the potential efficiencies of the four routes, we set several criteria including theoretical yield of IPDO, pathway thermodynamics estimated by the MDF, energy efficiency in form of ATP and NADPH cost, number of non-native steps, and penalties posed by inappropriate properties of intermediates and pathway enzymes (Table 1). We used FBA to predict their theoretical IPDO yields and corresponding metabolic flux distributions. Results showed that all three proposed IPDO pathways have higher IPDO theoretical yields than that of our previously reported route A (Table 1). The low yield (0.628 mol/mol glucose) of route A is due to the fact that hydroxylation of leucine into 4-OH-Leu in route A is coupled with the oxidation of  $\alpha$ -KG to succinate and reduction of the concentration of O<sub>2</sub> to CO<sub>2</sub>. This means that in addition to the carbon loss in the form of CO<sub>2</sub>, a certain metabolic flux has to be shunted into  $\alpha$ -KG formation to make this coupled reaction feasible, which may impose an "additional burden" for the efficient IPDO production via this route. In contrast, route C starting from  $\alpha,\beta$ -diOH-IV without carbon loss gives the highest IPDO theoretical yield, reaching 0.827 mol/mol glucose (Table 1 and Figure S2), which is 96.5% of the maximal theoretical yield (0.857 mol/mol glucose) determined purely by the energy balance.<sup>15</sup> The theoretical yields of IPDO of routes B reach 0.667 mol/mol glucose. Specifically, one carbon atom is lost as carbon dioxide (CO<sub>2</sub>) in the oxidative decarboxylation of  $\alpha$ -KIC to 3-OH-IV, which reduces its IPDO theoretical yield. As to route D, there are two reactions that involve the carbon loss: one is the conversion of pyruvate to acetyl-CoA, accompanied by CO<sub>2</sub> formation, and the other one is the last reaction step of converting 3,5-diOH-3-M-PA to IPDO by ADO, giving rise to a theoretical IPDO yield of 0.667 mol/mol glucose.



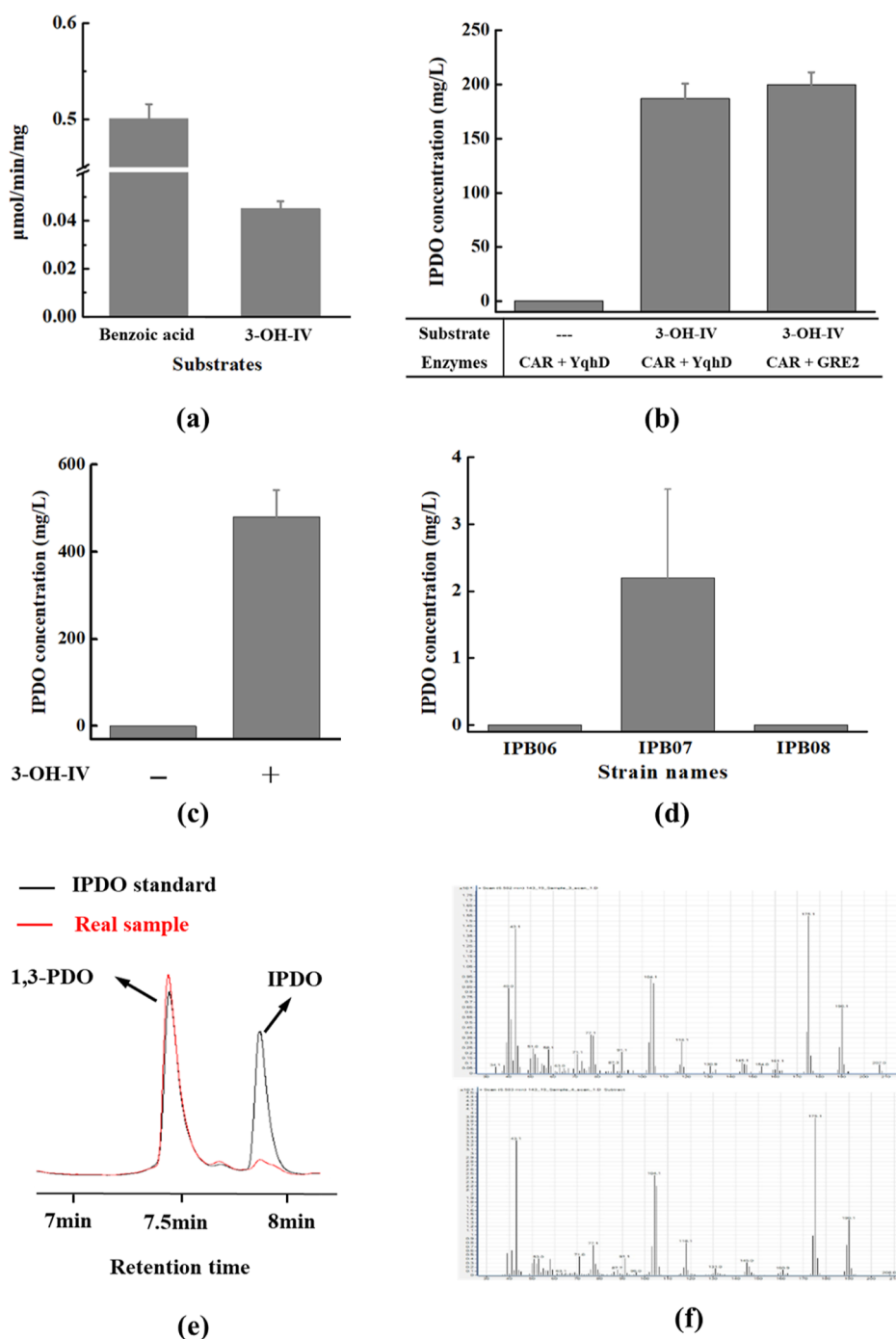
**Figure 2.** MDF values of four IPDO biosynthesis routes. Green dot lines represent the cumulative  $\Delta_r G'$  values of reactions in IPDO biosynthesis routes when the concentrations of all substrates are set as 1 mM, while solid lines represent the cumulative  $\Delta_r G'$  values of reactions when the concentrations of all substrates are under the optimal conditions evaluated by MDF. The reactions with the lowest  $\Delta_r G'$  values are regarded as bottleneck reactions (as shown in the red solid line in the figure). Note: <sup>a</sup>MDF values of all reactions in the IPDO synthesis routes. <sup>b</sup>MDF values of the rest of the reactions after removing the last common rate-limiting reaction in routes A and B.

In addition, the energy efficiencies of the four routes have been calculated. Since all four routes synthesize IPDO via extending pyruvate catabolism, we take pyruvate instead of glucose as the common precursor to recalculate the energy cost of each route. Among them, route A is the most energy efficient route, which only costs 2 molecular NAD(P)H to produce 1 molecular IPDO. Routes B and C exhibit equal energy efficiencies, requiring 2 molecular NAD(P)H in

addition to 1 molecular ATP to generate 1 molecular IPDO from pyruvate. The energy efficiency of route C is the lowest, which synthesizes 1 molecular IPDO from pyruvate with the input of 4 and 1 molecular NAD(P)H and ATP, respectively.

MDF is used to quantify a pathway's tendency to operate near-equilibrium and is calculated by setting the minimal driving forces of all reactions as the optimization goal and maximizing it.<sup>12</sup> MDF determines the magnitude of the





**Figure 3.** Verification of IPDO biosynthesis via route B. (a) Activity of CAR on 3-OH-IV. BA was used as the positive control. (b) Production of IPDO from 3-OH-IV by mutienzyme cascade reaction with purified CAR and YqhD or GRE2 in vitro. (c) Production of IPDO from 3-OH-IV in vivo. Strain IPB02 was cultivated in the FM-II medium with or without the addition of 50 mM 3-OH-IV into the FM-II medium. (d) IPDO production from glucose in vivo. (e) GC chromatograms of a calibration standard containing 100 mg/L of IPDO and a real sample from the IPB07 culture. 100 mg/L 1,3-PDO was used as the internal standard to monitor the derivatization efficiency with phenylboronic acid. (f) MS verification of IPDO production from 3-OH-IV as shown by the spectra of an IPDO standard (upper) and a real sample from IPB07 culture using glucose as the sole substrate (lower). 3-OH-IV, 3-hydroxyisovalerate. 1,3-PDO, 1,3-propanediol; IPDO, isopentylidol.

thermodynamic bottleneck of a given route. In principle, a route with a higher MDF value always means it has a less severe thermodynamic bottleneck and is thermodynamically more favorable. As also shown in Table 1, we determined the MDF values for all four IPDO synthetic routes under the given conditions (Section 4 in the “Materials and Methods”) to evaluate their thermodynamic favorability. Route D with an MDF value of 38.37 kJ/mol seems to be thermodynamically

most favorable (Table 1 and Figure 2) among the four routes. However, it has to be noted that its higher MDF value is mainly contributed by the high  $\Delta_r G'$  (−95.16 kJ/mol) of the last reaction catalyzed by ADO (Figures 1 and 2), which indicates that all upstream reactions in route D are rate-limiting reactions (Figure 2). This will practically be a problem for the subsequent optimization of this route. Route C gives the lowest MDF value (6.96 kJ/mol), which is mainly

attributed to the low  $\Delta_r G'$  values of the three reactions catalyzed by CAR ( $-0.74$  kJ/mol), PDDH ( $-1.26$  kJ/mol), and PDDH ( $-1.26$  kJ/mol) consecutively. In comparison, the MDF values of routes A and B are equal to and lower than that of route D. However, since routes A and B also share a thermodynamically clearly unfavorable reaction, that is, the last reaction catalyzed by PDDH ( $\Delta_r G'$ :  $-1.26$  kJ/mol, Figures 1 and 2), it seems that their MDF values are severely constrained by this bottleneck reaction. In this way, the contribution of other reactions to the thermodynamic driving forces of the routes is overwhelmingly obscured. To this end, we removed this common bottleneck reaction and recalculated the MDF values of the remaining reactions in routes A and B, which resulted in higher MDF values for both routes ( $84.67$  vs  $74.93$  kJ/mol). Moreover, the resulting cumulative  $\Delta_r G'$  values in routes A and B demonstrate a nearly linear decrease with each added reaction. The thermodynamically highly favorable reactions of hydroxylation, deamination, and decarboxylation in route A and the oxidative decarboxylation reaction in route B provide the powerful push needed for the thermodynamically unfavorable reduction formation of the hydroxyl group in the last step. This indicates that in pathway design, it is important to consider the thermodynamic topology of the reactions in an envisaged synthetic pathway.

The properties of intermediates and enzymes should also be taken into consideration to evaluate the potential of non-natural biosynthetic pathways. In many artificial biosynthetic pathways, some metabolites are produced as the outcome of the promiscuous activity of heterologous enzymes. The physiochemical properties of these metabolites may restrict the structure of the pathway and ultimately constrain the product yield. Major physiochemical properties of intermediates to be considered include toxicity, stability, and permeability.<sup>16</sup> In addition to high catalytic activity and specificity toward the target substrate, an ideal enzyme with industrial application prospects would have a clearly known catalytic mechanism, is easy and cheap to prepare and use, and is resistant to fluctuating cultivation environments (temperature, pH, etc.). Accordingly, we checked the intermediates and enzymes involved in the four IPDO routes and regarded the unfavorable intermediates and enzymes in these four routes as “penalties”. In this regard,  $\alpha$ -KIC and  $\alpha,\beta$ -DHIV, which are intermediates in routes B and C, respectively, exhibit lower affinity to HPD and CAR, respectively, compared to the endogenous enzymes that catalyze them, which makes it challenge to rewire the flux from the branched-chain amino acid biosynthesis pathways into the designed IPDO synthetic routes (Table 1). In addition, GDHt employed in route C requires the expensive vitamin B<sub>12</sub> as a cofactor to maintain its activity, which may largely limit its industrial application. The catalytic mechanism of aldehyde-deformylating oxygenase ADO in route D is still controversial,<sup>17–20</sup> making its subsequent rational engineering for further improvement of its activity challenging. The drawbacks of GDHt and ADO reduce the implementation priority of route C and route D.

Overall, we came up with three new IPDO biosynthetic routes and compared their potential efficiencies with our previously reported route A. All three new routes show higher theoretical IPDO yields than that of route A, implying greater potential from a glucose utilization perspective. Routes A and B, which produce IPDO by combining oxidative and reductive pathways of  $-OH$  group formation, require fewer reducing powers and exhibit higher thermodynamic driving force (as

shown by their higher MDF values and bottleneck reactions determined by MDF) than routes C and D that generate both hydroxyl groups of IPDO through the reductive pathway. In addition, route B gives the fewest non-native reaction steps among all four IPDO routes. Unlike routes C and D, route B does not employ enzymes whose activities are condition-dependent or not well-characterized. All of these advantages highlight the priority of route B.

In the follow-up study, we conducted the implementation of all three newly proposed IPDO biosynthetic routes in *E. coli* so as to explore the effectiveness of the methods and criteria used in this study for a priori in silico estimation of the thermodynamic feasibility of designed pathways for a given product and the relative likelihood of alternative reactions.

### 3. IMPLEMENTATION OF THE DESIGNED IPDO BIOSYNTHESIS ROUTES IN *E. COLI*

**3.1. Route B.** As illustrated in Figure 1, route B is designed to synthesize IPDO via extension of  $\alpha$ -KIC catabolism. HPD catalyzes the transformation of  $\alpha$ -KIC to 3-OH-IV, which is subsequently reduced to IPDO under the catalysis of CAR and PDDH, respectively. For the implementation, we began with the verification of the second step in route B, that is, the conversion of 3-OH-IV to 3-OH-3-M-BA catalyzed by CAR. This is because CAR has a wide scope of substrates due to its high promiscuous activity on non-native substrates.<sup>21</sup> To this end, CAR from *Nocardia iowensis* was selected as the first candidate because of its wide scope of substrates.<sup>21</sup> In addition, it has been reported that the 4'-phosphopantetheinyl transferases Sfp from *Bacillus subtilis* can catalyze post-translational modification of CAR by linking phosphopantetheine to the conserved serine of CAR to obtain a fully functional CAR.<sup>22</sup> Both the *car* and *sfp* genes were synthesized and inserted into pCDFDuet, a vector that is designed for the coexpression of two target genes driven by the T7 promoter, generating thereby the plasmid pCDF-car-sfp. The latter was transformed into the *E. coli* BL21(DE3) strain to obtain a recombinant strain named IPB01 for the subsequent expression and purification of CAR. The results indicate that coexpression of CAR and Sfp is crucial for enhancing the activity of CAR (Table S3), which is in accordance with the previous study.<sup>22</sup> For the enzyme activity assay of CAR on 3-OH-IV, benzoic acid (BA), the native substrate of CAR, was used as the positive control. As illustrated in Figure 3a, CAR showed activity on 3-OH-IV, although the activity was more 10 times lower than that on BA ( $0.045$  vs  $0.511$   $\mu\text{mol}/\text{min}/\text{mg}$ ).

Next, the feasibility of IPDO production from 3-OH-IV was verified both in vitro and in vivo. For the in vitro assay, the reaction was performed in HEPES buffer (pH 7.5) containing  $0$  or  $20$  mM 3-OH-IV,  $10$  mM NADPH,  $2$  mM ATP,  $10$  mM  $\text{Mg}^{2+}$ ,  $400$   $\mu\text{g}$  of CAR, and  $200$   $\mu\text{g}$  of either the ADH YqhD from *E. coli* or the methylglyoxal reductase GRE2 from *Saccharomyces cerevisiae* strain S288c. The reaction was incubated at  $30$  °C for  $6$  h. As shown in Figure 3b,  $187.2$  and  $200.0$  mg/L (nearly  $2$  mM) IPDO were formed from 3-OH-IV by the reactions containing YqhD or GRE2, respectively, while no IPDO was detected in the absence of 3-OH-IV. The higher IPDO titer of the reaction system employing GRE2 demonstrates that the exogenous GRE2 showed an even slightly higher activity on 3-OH-IV than the endogenous YqhD. Given that the amount of IPDO generated was close to the amount of ATP present in the reaction system, it is reasonable to claim that little intermediate 3-OH-3-M-BA

was accumulated, and the conversion of 3-OH-IV to 3-OH-3-M-BA by YqhD or GRE2 was very efficient.

To verify the feasibility of IPDO production from 3-OH-IV *in vivo*, we ligated GRE2 into pET28a to obtain pET-gre and cotransformed pET-gre and pCDF-car-sfp into *E. coli* BL21(DE3). The recombinant strain named IPB02 was inoculated into 5 mL of LB medium and grown overnight. The overnight culture was then inoculated into 10 mL of FM-II medium with an initial OD of 0.1 with or without the addition of 50 mM 3-OH-IV. 0.1 mM IPTG was added after OD<sub>600</sub> reached 0.4–0.6. The results showed that 480.0 mg/L IPDO was produced by IPB02 when 50 mM 3-OH-IV was supplemented in the medium, whereas no IPDO was detected in the absence of 3-OH-IV under the same conditions (Figure 3c).

Next, the synthesis of IPDO from  $\alpha$ -KIC was verified experimentally. We used whole-cell catalysis to convert  $\alpha$ -KIC into 3-OH-IV, as described in Section 7 of the [Materials and Methods](#) part. The obtained reaction mixture containing 3-OH-IV was subsequently used for the production of IPDO *in vitro*, as described above. As shown in Table 2, in the negative

**Table 2. Biosynthesis of IPDO from  $\alpha$ -KIC by Whole-Cell Catalysis Plus *In Vitro* Enzymatic Reactions**

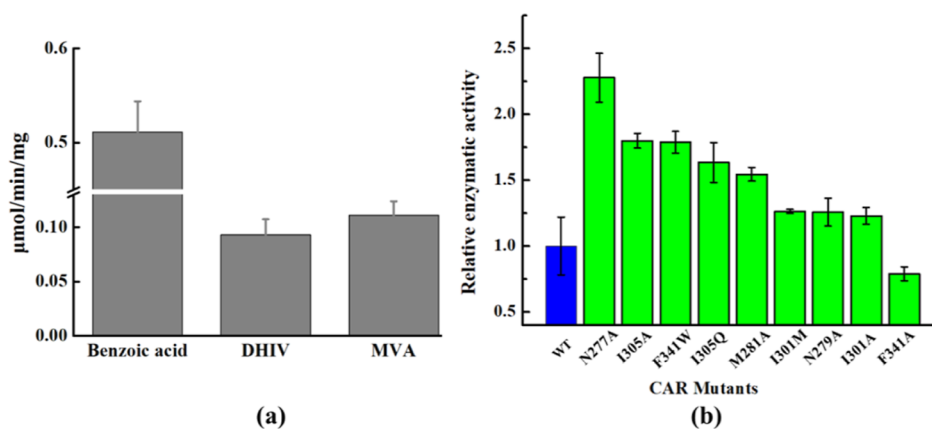
| strain            | production of 3-OH-IV from $\alpha$ -KIC by whole cell catalysis |          |                | in vitro conversion of 3-OH-IV to IPDO |                        |
|-------------------|------------------------------------------------------------------|----------|----------------|----------------------------------------|------------------------|
|                   | enzymes                                                          | time (h) | 3-OH-IV (mg/L) | 3-OH-IV (mg/L)                         | IPDO production (mg/L) |
| E. coli BL21(DE3) | no                                                               | 6        | N.D.           | N.D.                                   | N.D.                   |
| IPB03             | SnHPD                                                            | 6        | 914.3          | 274.3                                  | 232                    |
|                   |                                                                  | 24       | 1812.1         | 543.6                                  | 372                    |
| IPB04             | AaHPD                                                            | 6        | 1294.7         | 388.4                                  | 296                    |
|                   |                                                                  | 24       | 2119.2         | 635.8                                  | 416                    |

control using the *E. coli* BL21(DE3) strain, no conversion of  $\alpha$ -KIC to 3-OH-IV could be detected, while both SnHPD and AaHPD showed the  $\alpha$ -ketoisocaproate dioxygenase activity on the conversion of  $\alpha$ -KIC to 3-OH-IV, with AaHPD showing a slightly higher activity, as evidenced by the higher amounts of 3-OH-IV produced by IPB04 harboring AaHPD at both 6 and 24 h compared to those produced by IPB03 harboring SnHPD

(Table 2). In this regard, AhHPD was selected for further research of this pathway.

Based on the above results, we sought to explore the feasibility of producing IPDO *in vivo* from glucose. We first improved the CAR activity by optimizing the ratios of the expression levels of CAR and Sfp. The strain IPB05, which was constructed by integrating Sfp into the chromosome of *E. coli* W3110 driven by an artificial promoter J23105 (<http://parts.igem.org/>) and overexpressing CAR on the plasmid pTrc99a (pTrc-car), showed highest CAR activity (Table S3), so it was used for the subsequent verification of route B. Subsequently, GRE2 and AaHPD were inserted into the pTrc-car plasmid to obtain pTrc-cgh. After pTrc-car was substituted with pTrc-cgh in IPB05, the resulting strain (named IPB06) was cultivated in the FM-II medium for IPDO production. Unfortunately, no IPDO was detected to be produced by IPB06. One possible reason is that the supply of precursor  $\alpha$ -KIC is insufficient in IPB06. For this reason, we overexpressed the leucine operon (*leuA<sup>fb</sup>BCD<sup>23</sup>*) from *E. coli*, *alsS* from *B. subtilis*, and *ilvCD* genes from *E. coli* on the p15AS vector to obtain the plasmid p15-leu-acd and transformed p15-leu-acd into IPB06 to generate strain IPB07. Strain IPC08 obtained by transforming the empty p15AS vector into IPB06 was also constructed as the negative control. As a result, strain IPB07 produced 2.2 mg/L IPDO in a shake flask culture, while no IPDO was produced by the control strain IPB08 (Figure 3d–f), demonstrating that route B is functioning in *in vivo* biosynthesis of IPDO from glucose but needs to be improved in future studies.

**3.2. Routes C and D.** Route C starts from  $\alpha,\beta$ -DHIV and requires four exogenous reaction steps to produce IPDO. To begin with, the activity of CAR on  $\alpha,\beta$ -DHIV was tested *in vitro*. As illustrated in Figure 4a, the catalytic activity of CAR on  $\alpha,\beta$ -DHIV was successfully detected (0.093  $\mu\text{mol}/\text{min}/\text{mg}$ ) despite it being less than half of that on BA, the natural substrate of CAR. In order to perform *in vivo* validation of this route, plasmid pCOLA-yqhD-dhaB was constructed by ligating the *yqhD* gene encoding an ADH from *E. coli* and *dhaB* gene encoding GDHt from *Klebsiella* into the pCOLADuet vector, a vector with a ColA origin for the simultaneous coexpression of two genes. Then, the plasmids pCDF-car-sfp and pCOLA-yqhD-dhaB were cotransformed into *E. coli* BL21(DE3) to generate a recombinant strain named IPC01. Gas chromatog-



**Figure 4.** Implementation of IPDO biosynthesis via routes C and D in *E. coli*. (a) Verification of the catalytic activity of CAR on  $\alpha,\beta$ -DHIV and MVA *in vitro*. BA was taken as the positive control. (b) Relative activity of CAR mutants on MVA. All CAR mutants were obtained by mutating amino acid residues near the binding pocket or in the loop region outside the binding pocket.



raphy (GC) analysis showed that no IPDO production by IPC01 was detected after cultivation for 48 h of cultivation. Next, through a system metabolic engineering modification of IPC01, strain IPC02 was constructed to enhance the precursor supply by increasing the flux of the  $\alpha,\beta$ -DHIV biosynthesis pathway and block the branched-chain amino acid pathways in IPC01 (Figure S3). Unfortunately, strain IPC02 still did not produce IPDO.

Then, we turned to testing the feasibility of route C in vitro. All three enzymes CAR, DhaB, and YqhD were separately purified in vitro, and 25 mM  $\alpha,\beta$ -DHIV was used as the substrate. The reactions were carried out in two steps: the first reaction catalyzed by CAR was conducted for 1 h, and then YqhD and DhaB were added into the reaction system to continue the reactions for 3 h. Unfortunately, no IPDO was detected.

Routes A to C use either a branched-chain amino acid BCAA (leucine in route A) or intermediates ( $\alpha$ -KIC in route B and  $\alpha,\beta$ -DHIV in route C) in the BCAA biosynthesis pathway, therefore extending or diverting the BCAA metabolic fluxes to the synthesis of IPDO. As an alternative route, route D is quite different in using pyruvate as the precursor for IPDO biosynthesis. It requires three reaction steps to generate MVA from pyruvate and two reaction steps catalyzed successively by CAR and ADO to convert MVA to IPDO. For the implementation of route D in *E. coli*, we first verified that CAR shows activity on MVA (Figure 4a). Afterward, the *ado* gene encoding aldehyde-deformylating oxygenase from *Prochlorococcus marinus* as well as the *fdx* gene encoding ferredoxin from *Synechocystis* sp. 6803 and the *fmr* encoding ferredoxin (flavodoxin) NADP<sup>+</sup> reductase from *E. coli* that are used to supply NADPH for the ADO catalyzed reaction were inserted into the pCOLADuet vector to generate the plasmid pCOLA-aff. Then, the plasmids pCDF-car-sfp and pCOLA-aff were cotransformed into *E. coli* BL21(DE3) to obtain a recombinant strain named IPD01. IPD01 was cultivated in the FM-II medium by using glucose and MVA as cosubstrates. According to the GC analysis results, IPD01 did not generate IPDO after 48 h of cultivation even in the presence of MVA. One major reason for the failure of verification of routes C and D is attributed to their poor thermodynamic driving force, especially for the CAR catalyzed reaction and two reductases' catalyzed reactions in route C and the overall reactions from pyruvate to 3,5-diOH-3-M-PA in route D. In addition, the poor activities of CAR on  $\alpha,\beta$ -DHIV and MVA further reduce the efficiencies of routes C and D, rendering that no IPDO was formed or, at least, the generated IPDO was under the detection limit of GC analysis (less than 1 mg/L).

Next, rational engineering of CAR was carried out to improve its catalytic activity on  $\alpha,\beta$ -DHIV and MVA. Computational saturation mutagenesis/virtual screening<sup>24</sup> was conducted to predict CAR mutants with higher activities on MVA. Based on the crystal structure of CAR, we carried out rational protein design of CAR by focusing on either the binding pocket or on the loop near the binding pocket (Figure S4). Enzymatic activities of the predicted CAR mutants in comparison to those of wild-type CAR were examined in vitro, as shown in Figure 4b. It is found that most of the mutants exhibited increased activities toward MVA. Especially, the catalytic activity of the mutant N227A reached 2.28-fold of that of the wild type. Unfortunately, after these CAR mutants were introduced into route D instead of the wild-type CAR, none of the resulting recombinant *E. coli* strain appeared to be

able to produce IPDO in vivo. In addition to CAR, protein engineering was also performed on GDHt and ADO to improve their activities on the respective substrates (Figure S5). Unfortunately, none of these mutants led to the production of IPDO in vivo.

#### 4. DISCUSSION

In this study, we have designed three new IPDO biosynthesis pathways in addition to the biosynthetic route (named route A in this study) we reported before.<sup>7</sup> Among the three new routes, route B was computationally designed using Retropath RL and also successfully verified in *E. coli*, while no production of IPDO could be achieved via the two intuitively designed routes C and D. The feasibility of a synthetic pathway is controlled by many constraints, among which thermodynamics, kinetics, and regulation of the enzymes involved and physiochemical properties of metabolites (intermediates and products) and enzymes play predominant roles. In many artificial pathways, some non-natural intermediates are generated due to the promiscuity of the employed pathway enzymes, which makes it difficult to evaluate the physiochemical properties of these metabolites. The enzyme kinetic properties are always laborious and time-consuming to determine and often vary depending on the test conditions as well as on their origins, which largely limits their applications for describing the control that enzyme kinetics exerts on metabolic fluxes. As an alternative, thermodynamic analysis can be performed without using enzyme kinetic parameters and is not dependent on a particular initial steady state.<sup>12</sup> Here, taking route A as a reference route, we compared the four designed IPDO routes from the perspective of thermodynamic potential in order to better understand the implementation results. As mentioned above, the hydroxyl group can be obtained through either oxidation of an alkyl group (named the oxidation pathway) or reduction of a carboxyl group to a carbonyl group and then to a hydroxyl group (named the reduction pathway). The reactions that catalyze the formation of hydroxyl groups through an oxidation pathway are thermodynamically favorable since they are often coupled with other exergonic reactions. For instance, hydroxylation of  $\alpha$ -KIC by oxygenase HPD in route B couples the oxidation of  $\alpha$ -KIC with its decarboxylation, and hydroxylation of leucine by MFL in route A couples with the oxidation of  $\alpha$ -KG to succinate. However, the high energetics of these reactions ( $\Delta_r G'$ : -100 to -120 kcal/mol) comes at the expense of one carbon unit loss, which leads to the low theoretical yields of IPDO in routes A and B (Table 1). Nonetheless,  $\Delta_r G'$ -dependent thermodynamic analysis is not always decisive. For example, inorganic pyrophosphatase is a specific catalyst for the hydrolysis of pyrophosphate into orthophosphate. In this study, CAR is employed in routes B, C, and D to catalyze the reduction of the carboxylic acid to the corresponding aldehyde using NADPH as the electron donor, accompanied by the formation of diphosphate (PPi) (Figure 1). The subsequent hydrolysis of PPi provides a substantial additional thermodynamic driving force, thus making such reactions functionally irreversible.<sup>25</sup> However, the  $\Delta_r G'^0$  value of PPi hydrolysis is surprisingly positive (4.15 kcal/mol, data from MetaCyc EC 3.6.1.1), which means that the whole reaction is even thermodynamically unfavorable when the subsequent PPi hydrolysis is taken into consideration. We infer that this is mainly attributed to the differences in defined "standard conditions" and physiological environments. In



contrast to standard conditions, a previous study revealed that the  $\Delta_r G^0$  value of the hydrolysis of PPi reaction was estimated to be  $-16.74$  kJ/mol under simulated physiological conditions of pH and ionic strength, where the concentrations of Pi and PPi were  $2.42$  mM and  $6.2$   $\mu$ M, respectively.<sup>26</sup> The authors further concluded that the hydrolysis of the PPi reaction did not attain equilibrium in vivo. Therefore, it is unwise to merely use standard Gibson free energies of a reaction to simulate its thermodynamic driving force of cells under physiological states, especially when the working condition of a specific enzyme is vastly different from the defined “standard conditions.”

In addition to thermodynamic analysis, enzyme kinetics also plays a significant role in evaluating the feasibilities of artificial routes. Previous research studies reported that CARs have a broad substrate scope, which ranges from aliphatic acids to aromatic, polycyclic, and heterocyclic acids.<sup>27</sup> Although CARs attract significant interest, CARs only exhibit substantial reductase activity against substrates whose second functional group is separated from the carboxylate by at least three carbons,<sup>28</sup> which largely limit their applications in the biosynthesis field. Apparently, the intermediates  $\alpha,\beta$ -DHIV and MVA in routes C and D do not belong to CAR-preferred substrates, and therefore CAR exhibits poor catalytic activities on  $\alpha,\beta$ -DHIV and MVA, which is one of the key reasons that leads to the failure of implementation of routes C and D. Therefore, great importance should also be attached to the substrate preferences of the enzymes employed. Comparing the implementation results of the four designed pathways in *E. coli*, we infer that because of the high thermodynamic driving forces and enzymatic activities of routes A and B, a significant amount of metabolic flux is channeled to the IPDO synthesis pathway even at low concentrations of the precursors. In contrast, the failed implementation of routes C and D is mainly attributed to the poor kinetics of the reactions catalyzed by CAR. These results underline the significance of performing a priori thermodynamic and kinetic analysis to evaluate different biosynthetic pathways that produce the same product.

In addition, the efficiencies of the artificial pathways are influenced by the intrinsic environment of the host strain's intrinsic environment. These non-natural pathways are designed to synthesize target products by rewiring carbon flux from the inherent metabolisms in a competitive manner; therefore, the affinity and activity of the enzymes employed in a heterologous pathway for precursor intermediates are crucial for its efficiency. For instance, in route B,  $\alpha$ -KIC is the precursor for IPDO biosynthesis. In the wild-type *E. coli* strain, the majority of the endogenous  $\alpha$ -KIC pool is converted to leucine, which is catalyzed by the branched-chain-amino-acid transaminase IlvE at high efficiency ( $K_m$ :  $0.08$  mM,  $K_{cat}$ :  $24.7$  s<sup>-1</sup>).<sup>29</sup> In contrast, the catalytic ability of HPD on  $\alpha$ -KIC is much lower ( $K_m$ :  $0.32$  mM,  $K_{cat}$ :  $6.5$  min<sup>-1</sup>)<sup>30</sup> than that of IlvE. Given this, it is less efficient to use HPD to divert the endogenous  $\alpha$ -KIC from the leucine synthesis pathway into the IPDO synthesis pathway (Table 1). This is also the case for route C, in which CAR competes with endogenous dihydroxy-acid dehydratase IlvD for binding the substrate  $\alpha,\beta$ -DHIV for IPDO synthesis (Table 1). To further improve the efficiency of route B in future work, on one hand, the importance of improving  $\alpha$ -ketoisocaproate dioxygenase that exhibits improved activity on  $\alpha$ -KIC, which can be achieved through either rational protein engineering or screening its homologues from different species, should be highlighted. On the other

hand, a dynamic regulation system can be employed to dynamically control the expression abundance of IlvE. Recently, Hou et al. constructed a bifunctional molecular switch system by using growth phase-dependent promoters and degrons.<sup>31</sup> Within their system, dynamic regulation of the expression of IlvE in different growth phases can be fulfilled: a high expression level is kept at the exponential growth phase to ensure good cell growth but turned down to a relatively low level when cells reach the stationary phase to avoid shunting metabolic flux toward valine and leucine synthesis. In addition, a series of promoters with different strengths and induction phases have been characterized in their toolbox, which can be adopted in our future study for fine-tuning IlvE expression to maximize IPDO production.

## 5. MATERIALS AND METHODS

**5.1. Strains, Plasmids, Chemicals, and Cultivation Conditions.** Strains and plasmids used in this work are given in Table S1. *E. coli* Top10 was used for gene cloning and *E. coli* BL21(DE3) for protein expression. Primers used in this work are listed in Table S2. Genes encoding  $\alpha$ -ketoisocaproate dioxygenase, that is, SnHPD from *Streptomyces noursei* (DSM 40635) and AaHPD from *Allokutzneria albatra* (DSM 44149), and methylglyoxal reductase GRE2 from *S. cerevisiae* (strain S288c) were amplified from the corresponding genome DNA. Genes encoding CAR from *N. iowensis*, leucine hydroxylase MFL from *Methylobacillus flagellatus* KT, L-amino acid deaminase AAD from *Proteus vulgaris*, GDHt from *Klebsiella pneumoniae*, and aldehyde-deformylating oxygenase ADO from *Cyanobacteria* were synthesized by the Genscript Company (the Netherlands). Genes encoding  $\alpha$ -keto acid decarboxylase KDC from *Lactobacillus lactis* were previously synthesized in our lab.<sup>24</sup> MVA and IPDO were purchased from Sigma-Aldrich Company (Germany). 3-Hydroxyisovalerate (3-OH-IV),  $\alpha,\beta$ -DHIV, and  $\alpha$ -ketoisocaproate ( $\alpha$ -KIC) were purchased from the TCI Company (Japan). Isopropyl  $\beta$ -D-1-thiogalactopyranoside (IPTG) was purchased from Carl Roth Company (Germany). 4-Hydroxyleucine (4-OH-Leu) was purchased from AKos GmbH (Germany).

For the shake flask culture, individual colonies were picked up from agar plates and cultivated in LB medium (5 g/L yeast extract, 10 g/L tryptone, and 10 g/L NaCl) overnight. Harvested cells were then used to inoculate the fermentation medium II (FM-II medium containing 30 g/L glucose, 0.5 g/L MgSO<sub>4</sub>·7H<sub>2</sub>O, 3 g/L KH<sub>2</sub>PO<sub>4</sub>, 12 g/L K<sub>2</sub>HPO<sub>4</sub>, 4 g/L (NH<sub>4</sub>)<sub>2</sub>SO<sub>4</sub>, 1 g/L yeast extract, 2 g/L monosodium citrate, and 0.1 g/L FeSO<sub>4</sub>·7H<sub>2</sub>O, pH 7.0) at the inoculation rate of 1:100 (v/v). After OD<sub>600</sub> reached 0.4–0.6, 0.1 mM IPTG was added to induce protein expression, and the cultivation was carried out at 30 °C and 200 rpm for 48 h. Samples were collected by centrifugation, and the supernatants were filtered with a filter membrane (0.22  $\mu$ m) and stored at  $-20$  °C for subsequent analysis by GC or GC–mass spectrometry (GC–MS).

**5.2. Plasmid Construction.** Taking the pET-mfl as an example, the procedure of plasmid construction was as follows: the linearized *mfl* fragment with 15 bp extension homologous at its ends was obtained by using primers pET-mfl-LF and pET-mfl-LR with 50 ng of synthesized *mfl* as the template. The pET28a vector was linearized by using primers pET-vec-LF and pET-vec-LR with 50 ng of the pET28a plasmid as the template. The linearized pET28a vector and *mfl* fragment were ligated using the In-Fusion HD Cloning Kit (Clontech, Takara,

Japan), and the ligated product was then transformed into the *E. coli* Top10 strain and cultivated on an LB agar plate containing 50 mg/L kanamycin (Kan) at 37 °C overnight. The next day, colony PCR was applied to verify positive colonies, and the plasmids were extracted to sequence for further verification.

**5.3. Computational Pathway Design.** In order to design possible IPDO pathways, we performed a retrosynthesis analysis of IPDO using the software Retropath RL. First, we made each intermediate of the designed routes have a very high structural similarity to the structure of a corresponding natural intermediate metabolite. The substrate similarity cutoff value was set to 0.5, width to 10, depth to 7, max rollout to 10, reaction rule radius to 2, 4, 8, and 16, and score function to biological pathway scoring. The *E. coli* intermediate metabolite database available in the software was selected as Sinks, such that the retrosynthesis search would terminate upon reaching a metabolite synthesizable in *E. coli*. Then, to find more potential synthetic pathways, we varied the calculation parameters, reduced the substrate similarity to 0.1, switched to a random scoring method, and expanded the source of metabolites to all detectable intermediate metabolites.

**5.4. FBA of IPDO Biosynthesis via Four Different Artificial Routes.** The optimal metabolic flux distributions of the strains carrying different IPDO synthetic pathways were calculated using FBA and the genome-scale metabolic model iY75\_1357 for *E. coli* str. K-12 substr. W3110 ([http://bigg.ucsd.edu/models/iY75\\_1357](http://bigg.ucsd.edu/models/iY75_1357)). The reactions in each artificial route were as follows:

Route A

- (1)  $\text{Leu} + \alpha\text{-KG} + \text{O}_2 \rightarrow 4\text{-OH-Leu} + \text{succinate} + \text{CO}_2$
- (2)  $4\text{-OH-Leu} + 1/2 \text{O}_2 \rightarrow 4\text{-OH-}\alpha\text{-KIC} + \text{NH}_3$
- (3)  $4\text{-OH-}\alpha\text{-KIC} \rightarrow 3\text{-OH-3-M-BA} + \text{CO}_2$
- (4)  $3\text{-OH-3-M-BA} + \text{NADPH} + \text{H}^+ \rightarrow \text{IPDO} + \text{NADP}^+$

Route B

- (1)  $\alpha\text{-KIC} + \text{O}_2 \rightarrow 3\text{-OH-IV} + \text{CO}_2$
- (2)  $3\text{-OH-IV} + \text{ATP} + \text{NADPH} + \text{H}^+ \rightarrow 3\text{-OH-3-M-BA} + \text{AMP} + \text{Ppi} + \text{NADP}^+$
- (3)  $3\text{-OH-3-M-BA} + \text{NADPH} + \text{H}^+ \rightarrow \text{IPDO} + \text{NADP}^+$

Route C

- (1)  $\alpha,\beta\text{-DHIV} + \text{ATP} + \text{NADPH} + \text{H}^+ \rightarrow 2,3\text{-diOH-IVA} + \text{AMP} + \text{Ppi} + \text{NADP}^+$
- (2)  $2,3\text{-diOH-IVA} + \text{NADPH} + \text{H}^+ \rightarrow 3\text{-M-BTO} + \text{NADP}^+$
- (3)  $3\text{-M-BTO} \rightarrow 3\text{-OH-3-M-BA} + \text{H}_2\text{O}$
- (4)  $3\text{-OH-3-M-BA} + \text{NADPH} + \text{H}^+ \rightarrow \text{IPDO} + \text{NADP}^+$

Route D

- (1)  $\text{acetoacetyl-CoA} + \text{acetyl-CoA} + \text{H}_2\text{O} \rightarrow \text{HMG-CoA} + \text{CoA}$
- (2)  $\text{HMG-CoA} + 2\text{NADPH} + 2\text{H}^+ \rightarrow \text{MVA} + \text{CoA} + 2\text{NADP}^+$
- (3)  $\text{MVA} + \text{ATP} + \text{NADPH} + \text{H}^+ \rightarrow 3,5\text{-diOH-3-M-PA} + \text{AMP} + \text{Ppi} + \text{NADP}^+$
- (4)  $3,5\text{-diOH-3-M-PA} + 2\text{NADPH} + \text{O}_2 + 2\text{H}^+ \rightarrow \text{IPDO} + \text{HCOOH} + \text{H}_2\text{O} + 2\text{NADP}^+$

The resulting metabolic submodels were named as iY75\_1357\_IPDO\_A, iY75\_1357\_IPDO\_B, iY75\_1357\_IPDO\_C, and iY75\_1357\_IPDO\_D and used for a constraint-based FBA, in which the glucose uptake rate was set at 15 mmol/g/DCW and the sources of oxygen, nitrogen, phosphate, and sulfur were not constrained.

**5.5. Thermodynamic Analysis of the Four IPDO Synthetic Routes.** For analyzing the thermodynamic efficiency of the four IPDO synthetic routes, Gibbs free energies of reactions ( $\Delta_r G^{\circ}$ ) were calculated using the component contribution method.<sup>32</sup> The MDF analysis was performed using the Python packages equilibrator-api and equilibrator pathway.<sup>33</sup> The concentration boundaries of metabolites were set with reference to the literature reports<sup>34</sup> as follows: ATP: 1 mM; ADP: 0.1 mM; Pi: 0.1 mM; NADPH: 1 mM; NADP: 0.1 mM; AcCoA: 10 mM; AaCoA: 0.0005–10 mM; CoA: 0.0005–10 mM; IPDO: 0.0005–10 mM; 3HmCoA: 0.0005–10 mM; HS\_CoA: 0.0005–10 mM; MVA: 0.0005–10 mM; 35dmp: 0.0005–10 mM; formate: 0.0005–10 mM.

**5.6. Enzyme Expression and Purification.** For the expression and purification of each target protein, a single colony from the LB agar plate was inoculated into 5 mL of LB medium at 37 °C and 220 rpm overnight. The overnight culture was then inoculated into 50 mL of LB medium in 300 mL shake flasks. After growing at 37 °C, 220 rpm to an OD<sub>600</sub> of 0.4–0.6, 0.1 mM IPTG was added into the medium to induce the enzyme expression, and the cultivation was continued at 30 °C and 220 rpm for 12–16 h. After cooling down on ice for 30 min, cells were harvested by centrifugation at 4 °C and 5000 rpm and washed with 30 mL of a buffer containing 20 mM sodium phosphate, 500 mM NaCl, and 20 mM imidazole (pH 7.4). Cell pellets were resuspended in 3 mL of the same buffer and disrupted by the multidirectional, simultaneous beating with specialized lysing matrix beads in a FastPrep 24 homogenizer (MP Biomedicals). The supernatant containing the target protein was collected by centrifuging at 4 °C and 13,000 rpm for 20 min. The target protein was purified by loading the supernatant onto a prepacked His SpinTrap column (GE Healthcare) and eluting with a buffer containing 20 mM sodium phosphate, 500 mM NaCl, and 500 mM imidazole (pH 7.4), followed by buffer change into a HEPES buffer (pH 7.5) using the Amicon Ultra-0.5 centrifugal filter device (10K MWCO) at 4 °C for the subsequent enzyme assay. The concentration of the purified protein was determined according to Bradford's method using bovine serum albumin as the standard.

**5.7. Enzymatic Activity Assays.** Assay of the CAR activity was performed as described previously<sup>35</sup> with minor modifications. The enzyme activity was measured by monitoring the disappearance of NADPH via the absorbance at 340 nm. The reaction mixture was composed of 80 mM HEPES buffer (pH 7.5), 10 mM MgCl<sub>2</sub>, 0.25 mM NADPH, 1 mM ATP, 5 mM of a carboxylic acid substrate, and 100  $\mu\text{g}$  of purified CAR in a total volume of 0.2 mL. The reaction mixture without the substrate was incubated to 30 °C, and the reaction was started by adding the corresponding carboxylic acid and allowed to carry on for 5–15 min. The enzymatic activity was determined by monitoring the disappearance of NADPH and measuring the absorbance at 340 nm.

**5.8. Whole-Cell Catalyzed Conversion of  $\alpha$ -KIC to 3-OH-IV.** For HPD expression, corresponding genes of two  $\alpha$ -ketoisocaproate dioxygenase (HPD), namely, SnHPD from *S. noursei* and AaHPD from *A. albatra*, were separately inserted into the pET22b vector and transformed into *E. coli* BL21(DE3) to yield recombinant strains IPB03 and IPB04, respectively. IPB03 and IPB04 were cultivated in LB medium containing 0.1 mM IPTG to induce the expression of the HPDs. After cultivating overnight, cells were harvested by

centrifugation at 4 °C and 5000 rpm for 10 min, washed with 0.9% NaCl, and then resuspended in 5 mL of 0.1 M Tris-buffer (pH 6.5) containing 33 mM  $\alpha$ -KIC, 3 mM FeSO<sub>4</sub>, 1 mM DTT, and 5 mM ascorbate. The whole-cell catalysis of  $\alpha$ -KIC conversion to 3-OH-IV was carried out at 30 °C and 220 rpm in a 100 mL baffled shake flask. Samples were taken after 6 and 24 h of incubation to measure the concentration of remaining 3-OH-IV in the supernatant.

**5.9. Analysis of IPDO by GC and GC–MS.** To measure the amount of IPDO in fermentation broth, 50  $\mu$ L of the filtered culture supernatant was mixed with 100  $\mu$ L of acetonitrile and 50  $\mu$ L of a derivatization reagent (comprising 5 mg/mL phenylboronic acid dissolved in 2,2-dimethoxypropane) in a 1.5 mL tube, and the derivatization was performed by shaking at 600 rpm at room temperature for 15 min. After centrifugation to remove the precipitated protein, the supernatant was transferred into a glass vial for subsequent GC analysis.

GC analysis of the derivatized IPDO was achieved on a Phenomenex HP-5 capillary column (30 m  $\times$  0.25 mm  $\times$  0.25  $\mu$ m) using a Varian 3900 GC instrument equipped with a flame ionization detector. 1  $\mu$ L of the derivatized sample was injected at 270 °C using a splitless/split injector (splitless for 1 min and then split at a ratio of 1:10). Nitrogen gas was used as the carrier gas at a flow rate of 1.5 mL/min. The column oven temperature program was as follows: held at 100 °C for 2 min, then ramped at 10 °C/min to 220 °C, and finally ramped at 20 °C/min to 260 °C and held for 5 min. The flame ionization detector was operated at 300 °C.

GC–MS analysis of the derivatized IPDO was carried out using an Agilent 7890B gas chromatograph equipped with an Agilent DB-5 ms UI nonpolar capillary column (30 m  $\times$  250  $\mu$ m  $\times$  0.25  $\mu$ m) and coupled to an Agilent MSD 5977 mass spectrometer. 0.2  $\mu$ L of the sample was injected into a Gerstel CIS injector at a split ratio of 1:5. The initial operating temperature was 60 °C (hold for 0.1 min) and then ramped at a rate of 12 °C/s to 270 °C and held for 1 min. The column oven was held at an initial temperature of 60 °C for 2 min and then ramped to 270 °C at 20 °C/min and held for 12 min. The MS was performed in scan mode in the range of 2–500 amu with a solvent delay of 4.5 min. The confirmation of IPDO was performed through a comparison of the mass spectra acquired as well as the retention times between the sample and the IPDO standard.

## ■ ASSOCIATED CONTENT

### SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssynbio.3c00394>.

Two IPDO biosynthesis routes designed using the software Retropath RL; optimal metabolic flux distribution for IPDO production by four different artificial routes; system metabolic engineering on *E. coli* BL21-(DE3) to enhance the accumulation of  $\alpha,\beta$ -DHIV; rational protein design of CAR; rational protein design of GDHt and ADO; strains and plasmids used in this work; primers used in this work; and comparison of the activities of CAR overexpressed from different hosts (PDF)

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<https://pubs.acs.org/doi/10.1021/acssynbio.3c00394>

### Author Contributions

Y.L. and L.C. contributed equally to this article. A.-P.Z., Y.L., L.C., and C.M. conceived the concept of the work. Y.L., L.C., C.M., and Q.Y. designed and performed the experimental studies. Y.L., L.C., and W.W. did the analytic work. Y.L., L.C., W.W., Q.Y., H.M., C.Z., C.M., and A.Z. were involved in data analysis and discussions and in writing the manuscript. A.-P.Z. provided the financial support and supervised the work.

### Notes

The authors declare no competing financial interest.

## ■ ACKNOWLEDGMENTS

We thank Hua An Tang Company (China) for providing financial support for this study. Y.L. is supported by a PhD scholarship from the Chinese Scholarship Council (CSC), which is gratefully acknowledged.



## REFERENCES

- (1) Mitsunashi, S. Current topics in the biotechnological production of essential amino acids, functional amino acids, and dipeptides. *Curr. Opin. Biotechnol.* **2014**, *26*, 38–44.
- (2) Lee, S. Y.; Kim, H. U. Systems strategies for developing industrial microbial strains. *Nat. Biotechnol.* **2015**, *33*, 1061–1072.
- (3) Finnigan, W.; Hepworth, L. J.; Flitsch, S. L.; Turner, N. J. RetroBioCat as a computer-aided synthesis planning tool for biocatalytic reactions and cascades. *Nat. Catal.* **2021**, *4*, 98–104.
- (4) Zhang, Y.; Liu, D.; Chen, Z. Production of C2–C4 diols from renewable bioresources: new metabolic pathways and metabolic engineering strategies. *Biotechnol. Biofuels* **2017**, *10*, 299.
- (5) Zeng, A. P.; Sabra, W. Microbial production of diols as platform chemicals: recent progresses. *Curr. Opin. Biotechnol.* **2011**, *22*, 749–757.
- (6) Wang, J.; Li, C.; Zou, Y.; Yan, Y. Bacterial synthesis of C3 C5 diols via extending amino acid catabolism. *Proc. Natl. Acad. Sci. U.S.A.* **2020**, *117*, 19159–19167.
- (7) Liu, Y. W. W.; Zeng, A.-p. Biosynthesizing structurally diverse diols via a general route combining oxidative and reductive formations of OH-groups. **2022**.
- (8) Hadadi, N.; Hatzimanikatis, V. Design of computational retrobiosynthesis tools for the design of de novo synthetic pathways. *Curr. Opin. Chem. Biol.* **2015**, *28*, 99–104.
- (9) Carbonell, P.; Parutto, P.; Baudier, C.; Junot, C.; Faulon, J. L. Retropath: automated pipeline for embedded metabolic circuits. *ACS Synth. Biol.* **2014**, *3*, S65–S77.
- (10) Yim, H.; Haselbeck, R.; Niu, W.; Pujol-Baxley, C.; Burgard, A.; Boldt, J.; Khandurina, J.; Trawick, J. D.; Osterhout, R. E.; Stephen, R.; et al. Metabolic engineering of *Escherichia coli* for direct production of 1,4-butanediol. *Nat. Chem. Biol.* **2011**, *7*, 445–452.
- (11) Orth, J. D.; Thiele, I.; Palsson, B. O. What is flux balance analysis? *Nat. Biotechnol.* **2010**, *28*, 245–248.
- (12) Noor, E.; Bar-Even, A.; Flamholz, A.; Reznik, E.; Liebermeister, W.; Milo, R. Pathway thermodynamics highlights kinetic obstacles in central metabolism. *PLoS Comput. Biol.* **2014**, *10*, No. e1003483.
- (13) Ye, L.; He, P.; Li, Q.; Zhang, X.; Bi, C. Type II restriction based combinatory modulation technique for metabolic pathway optimization. *Microb. Cell Factories* **2017**, *16*, 47.
- (14) Wang, J. L.; Niyompanich, S.; Tai, Y. S.; Wang, J.; Bai, W.; Mahida, P.; Gao, T.; Zhang, K. Engineering of a highly efficient *Escherichia coli* strain for mevalonate fermentation through chromosomal integration. *Appl. Environ. Microbiol.* **2016**, *82*, 7176–7184.
- (15) Dugar, D.; Stephanopoulos, G. Relative potential of biosynthetic pathways for biofuels and bio-based products. *Nat. Biotechnol.* **2011**, *29*, 1074–1078.
- (16) Bar-Even, A.; Flamholz, A.; Noor, E.; Milo, R. Rethinking glycolysis: on the biochemical logic of metabolic pathways. *Nat. Chem. Biol.* **2012**, *8*, S09–S17.
- (17) Jia, C.; Li, M.; Li, J.; Zhang, J.; Zhang, H.; Cao, P.; Pan, X.; Lu, X.; Chang, W. Structural insights into the catalytic mechanism of aldehyde-deformylating oxygenases. *Protein Cell* **2015**, *6*, 55–67.
- (18) Pandelia, M. E.; Li, N.; Norgaard, H.; Warui, D. M.; Rajakovich, L. J.; Chang, W. c.; Booker, S. J.; Krebs, C.; Bollinger, J. M. Substrate-triggered addition of dioxygen to the diferric cofactor of aldehyde-deformylating oxygenase to form a diferric-peroxide intermediate. *J. Am. Chem. Soc.* **2013**, *135*, 15801–15812.
- (19) Aukema, K. G.; Makris, T. M.; Stoian, S. A.; Richman, J. E.; Münck, E.; Lipscomb, J. D.; Wackett, L. P. Cyanobacterial aldehyde deformylase oxygenation of aldehydes yields n-1 aldehydes and alcohols in addition to alkanes. *ACS Catal.* **2013**, *3*, 2228–2238.
- (20) Wang, C.; Zhao, C. Y.; Hu, L. R.; Chen, H. Calculated mechanism of cyanobacterial aldehyde-deformylating oxygenase: asymmetric aldehyde activation by a symmetric diiron cofactor. *J. Phys. Chem. Lett.* **2016**, *7*, 4427–4432.
- (21) Winkler, M. Carboxylic acid reductase enzymes (CARs). *Curr. Opin. Chem. Biol.* **2018**, *43*, 23–29.
- (22) Finnigan, W.; Thomas, A.; Cromar, H.; Gough, B.; Snajdrova, R.; Adams, J. P.; Littlechild, J. A.; Harmer, N. J. Characterization of carboxylic acid reductases as enzymes in the toolbox for synthetic chemistry. *ChemCatChem* **2017**, *9*, 1005–1017.
- (23) Tashiro, Y.; Rodriguez, G. M.; Atsumi, S. 2-Keto acids based biosynthesis pathways for renewable fuels and chemicals. *J. Ind. Microbiol. Biotechnol.* **2015**, *42*, 361–373.
- (24) Zhang, Y. J.; Ma, C. W.; Dischert, W.; Soucaille, P.; Zeng, A. Engineering of phosphoserine aminotransferase increases the conversion of L-homoserine to 4-hydroxy-2-ketobutyrate in a glycerol-independent pathway of 1,3-propanediol production from glucose. *Biotechnol. J.* **2019**, *14*, No. e1900003.
- (25) Russell, R. G. G. Metabolism of Inorganic Pyrophosphate (Ppi). *Arthritis Rheum.* **1976**, *19*, 465–478.
- (26) Flodgaard, H.; Fleron, P. Thermodynamic Parameters for the Hydrolysis of Inorganic Pyrophosphate at pH 7.4 as a Function of [Mg<sup>2+</sup>], [K<sup>+</sup>], and Ionic Strength Determined from Equilibrium Studies of the Reaction. *J. Biol. Chem.* **1974**, *249*, 3465–3474.
- (27) Winkler, M. Carboxylic acid reductase enzymes (CARs). *Curr. Opin. Chem. Biol.* **2018**, *43*, 23–29.
- (28) Khusnutdinova, A. N.; Flick, R.; Popovic, A.; Brown, G.; Tchigvintsev, A.; Nocek, B.; Correia, K.; Joo, J. C.; Mahadevan, R.; Yakunin, A. F. Exploring Bacterial Carboxylate Reductases for the Reduction of Bifunctional Carboxylic Acids. *Biotechnol. J.* **2017**, *12*, 1600751.
- (29) Yu, X. J.; Wang, X. G.; Engel, P. C. The specificity and kinetic mechanism of branched-chain amino acid aminotransferase from *Escherichia coli* studied with a new improved coupled assay procedure and the enzyme's potential for biocatalysis. *FEBS J.* **2014**, *281*, 391–400.
- (30) Sabourin, P. J.; Bieber, L. L. Purification and characterization of an alpha-ketoisocaproate oxygenase of rat liver. *J. Biol. Chem.* **1982**, *257*, 7460–7467.
- (31) Hou, J. S.; Gao, C.; Guo, L.; Nielsen, J.; Ding, Q.; Tang, W.; Hu, G.; Chen, X.; Liu, L. Rewiring carbon flux in *Escherichia coli* using a bifunctional molecular switch. *Metab. Eng.* **2020**, *61*, 47–57.
- (32) Jankowski, M. D.; Henry, C. S.; Broadbelt, L. J.; Hatzimanikatis, V. Group contribution method for thermodynamic analysis of complex metabolic networks. *Biophys. J.* **2008**, *95*, 1487–1499.
- (33) Flamholz, A.; Noor, E.; Bar-Even, A.; Milo, R. eQuilibrator—the biochemical thermodynamics calculator. *Nucleic Acids Res.* **2012**, *40*, D770–D775.
- (34) Yang, X.; Mao, Z.; Zhao, X.; Wang, R.; Zhang, P.; Cai, J.; Xue, C.; Ma, H. Integrating thermodynamic and enzymatic constraints into genome-scale metabolic models. *Metab. Eng.* **2021**, *67*, 133–144.
- (35) Moura, M.; Pertusi, D.; Lenzini, S.; Bhan, N.; Broadbelt, L. J.; Tyo, K. E. Characterizing and predicting carboxylic acid reductase activity for diversifying bioaldehyde production. *Biotechnol. Bioeng.* **2016**, *113*, 944–952.